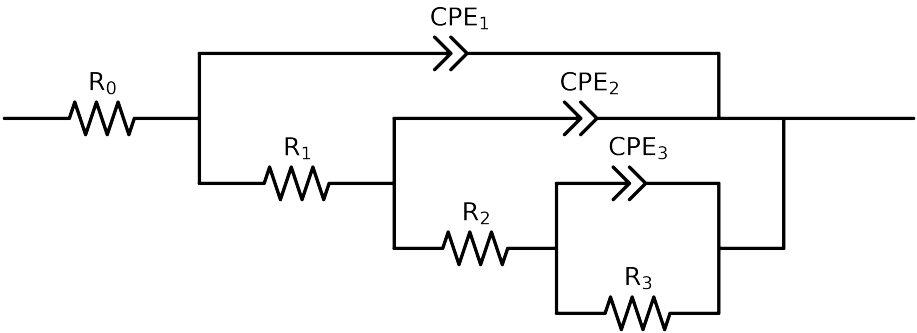


Comparative EIS Kinetics Report

Model Structure: $R_0-p(R_1-p(R_2-p(R_3,CPE_3),CPE_2),CPE_1)$

Used data: original data



Element	Unit	Bounds	Vasco_ LNV3_ 1	Vasco_ LNV3_ 3
R_0	Ω	$[1.0e+00, 1.0e+03]$	$1.38e+02 \pm 1.5e+00$	$1.39e+02 \pm 1.0e+00$
R_1	Ω	$[1.0e+00, 1.0e+08]$	$1.49e+03 \pm 2.4e+02$	$1.15e+03 \pm 1.4e+02$
R_2	Ω	$[1.0e+00, 1.0e+08]$	$2.92e+03 \pm 7.2e+02$	$2.81e+03 \pm 8.0e+02$
R_3	Ω	$[1.0e+00, 1.0e+08]$	$1.48e+06 \pm 1.2e+05$	$1.16e+06 \pm 3.2e+04$
Q_3	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$1.97e-06 \pm 4.7e-07$	$1.67e-06 \pm 5.4e-07$
n_3	-	$[5.0e-01, 1.0e+00]$	$9.50e-01 \pm 2.9e-02$	$9.50e-01 \pm 3.8e-02$
Q_2	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-02]$	$8.06e-07 \pm 6.8e-07$	$9.60e-07 \pm 7.6e-07$
n_2	-	$[5.0e-01, 1.0e+00]$	$9.24e-01 \pm 1.2e-01$	$9.22e-01 \pm 1.0e-01$
Q_1	$\Omega^{-1} \text{ sec}^a$	$[1.0e-09, 1.0e-03]$	$1.79e-06 \pm 3.0e-07$	$2.47e-06 \pm 2.6e-07$
n_1	-	$[5.0e-01, 1.0e+00]$	$7.56e-01 \pm 1.5e-02$	$7.37e-01 \pm 9.2e-03$
χ^2	-	-	$2.860e-04$	$2.350e-04$

Analysis Results: Vasco_LNV3_1

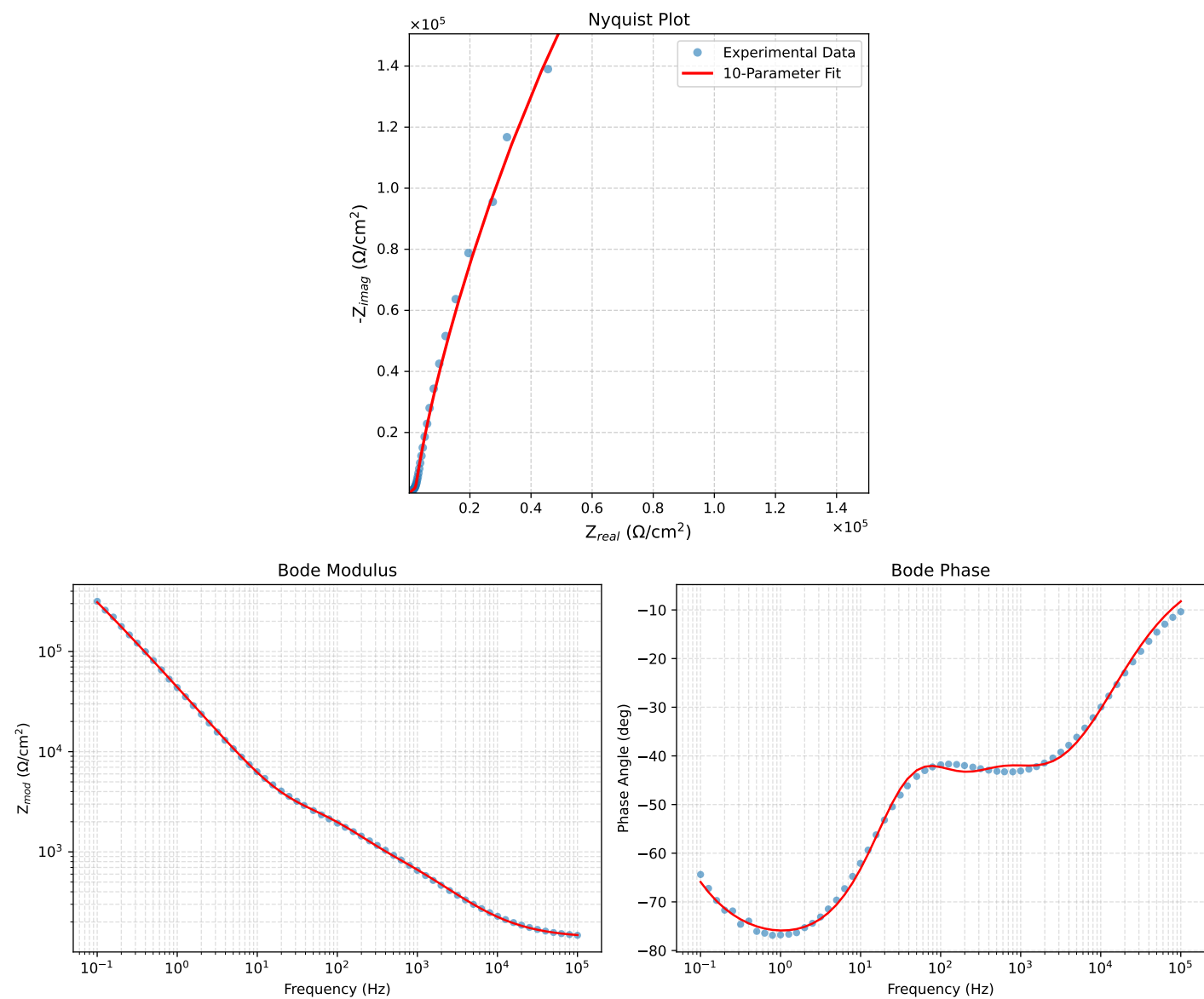


Figure 1: EIS Fit Results for Vasco_LNV3_1

Fitted Data: Vasco_LNV3_1

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0008e+05	1.4646e+02	2.1185e+01	1.4799e+02	-8.2303
7.9453e+04	1.4821e+02	2.5164e+01	1.5034e+02	-9.6360
6.3141e+04	1.5031e+02	2.9856e+01	1.5325e+02	-11.2343
5.0203e+04	1.5283e+02	3.5390e+01	1.5688e+02	-13.0374
3.9891e+04	1.5588e+02	4.1941e+01	1.6143e+02	-15.0590
3.1641e+04	1.5960e+02	4.9729e+01	1.6717e+02	-17.3062
2.5172e+04	1.6405e+02	5.8777e+01	1.7426e+02	-19.7121
2.0016e+04	1.6945e+02	6.9413e+01	1.8312e+02	-22.2754
1.5891e+04	1.7607e+02	8.1947e+01	1.9421e+02	-24.9578
1.2609e+04	1.8416e+02	9.6603e+01	2.0796e+02	-27.6792
1.0078e+04	1.9371e+02	1.1306e+02	2.2429e+02	-30.2703
8.0156e+03	2.0557e+02	1.3242e+02	2.4453e+02	-32.7885
6.3281e+03	2.2055e+02	1.5537e+02	2.6978e+02	-35.1642
5.0156e+03	2.3853e+02	1.8107e+02	2.9947e+02	-37.2026
3.9844e+03	2.6012e+02	2.0970e+02	3.3412e+02	-38.8748
3.1710e+03	2.8585e+02	2.4124e+02	3.7404e+02	-40.1625
2.5276e+03	3.1616e+02	2.7553e+02	4.1938e+02	-41.0724
1.9761e+03	3.5486e+02	3.1588e+02	4.7509e+02	-41.6743
1.5775e+03	3.9571e+02	3.5545e+02	5.3191e+02	-41.9326
1.2656e+03	4.4049e+02	3.9656e+02	5.9270e+02	-41.9960
9.9826e+02	4.9366e+02	4.4393e+02	6.6391e+02	-41.9641
7.9688e+02	5.4845e+02	4.9289e+02	7.3738e+02	-41.9458
6.2779e+02	6.1101e+02	5.5079e+02	8.2262e+02	-42.0328
5.0551e+02	6.7265e+02	6.1084e+02	9.0861e+02	-42.2427
3.9800e+02	7.4802e+02	6.8760e+02	1.0160e+03	-42.5897
3.1550e+02	8.3160e+02	7.7409e+02	1.1361e+03	-42.9488
2.5240e+02	9.2471e+02	8.6830e+02	1.2685e+03	-43.1980
1.9862e+02	1.0415e+03	9.7990e+02	1.4300e+03	-43.2549
1.5836e+02	1.1687e+03	1.0926e+03	1.5999e+03	-43.0723
1.2556e+02	1.3139e+03	1.2126e+03	1.7880e+03	-42.7025
1.0045e+02	1.4629e+03	1.3315e+03	1.9782e+03	-42.3084
7.9003e+01	1.6258e+03	1.4670e+03	2.1898e+03	-42.0604
6.3345e+01	1.7707e+03	1.6062e+03	2.3907e+03	-42.2104
5.0223e+01	1.9126e+03	1.7810e+03	2.6134e+03	-42.9592
3.8422e+01	2.0603e+03	2.0418e+03	2.9007e+03	-44.7412
3.1250e+01	2.1642e+03	2.3051e+03	3.1619e+03	-46.8051
2.4934e+01	2.2722e+03	2.6748e+03	3.5096e+03	-49.6525
1.9862e+01	2.3815e+03	3.1554e+03	3.9533e+03	-52.9573
1.5625e+01	2.5061e+03	3.8082e+03	4.5589e+03	-56.6517
1.2401e+01	2.6450e+03	4.6096e+03	5.3146e+03	-60.1534
9.9311e+00	2.8057e+03	5.5735e+03	6.2399e+03	-63.2796
7.9449e+00	3.0052e+03	6.7729e+03	7.4097e+03	-66.0727
6.3174e+00	3.2634e+03	8.2960e+03	8.9148e+03	-68.5269
5.0080e+00	3.5972e+03	1.0204e+04	1.0819e+04	-70.5810
3.9457e+00	4.0383e+03	1.2628e+04	1.3258e+04	-72.2657
3.1587e+00	4.5662e+03	1.5406e+04	1.6069e+04	-73.4909
2.5040e+00	5.2719e+03	1.8955e+04	1.9675e+04	-74.4575
1.9981e+00	6.1521e+03	2.3172e+04	2.3975e+04	-75.1312
1.5847e+00	7.3143e+03	2.8459e+04	2.9384e+04	-75.5863
1.2669e+00	8.7579e+03	3.4675e+04	3.5764e+04	-75.8251
9.9904e-01	1.0742e+04	4.2715e+04	4.4045e+04	-75.8847
7.9234e-01	1.3266e+04	5.2291e+04	5.3948e+04	-75.7642
6.3345e-01	1.6433e+04	6.3484e+04	6.5576e+04	-75.4870
5.0403e-01	2.0644e+04	7.7264e+04	7.9974e+04	-75.0407
4.0064e-01	2.6192e+04	9.3937e+04	9.7520e+04	-74.4202
3.1672e-01	3.3702e+04	1.1446e+05	1.1932e+05	-73.5932
2.5202e-01	4.3382e+04	1.3830e+05	1.4495e+05	-72.5846
2.0032e-01	5.6261e+04	1.6664e+05	1.7588e+05	-71.3446
1.5890e-01	7.3521e+04	2.0018e+05	2.1326e+05	-69.8334
1.2601e-01	9.6466e+04	2.3907e+05	2.5780e+05	-68.0255
1.0016e-01	1.2644e+05	2.8280e+05	3.0978e+05	-65.9102

Analysis Results: Vasco_LNV3_3

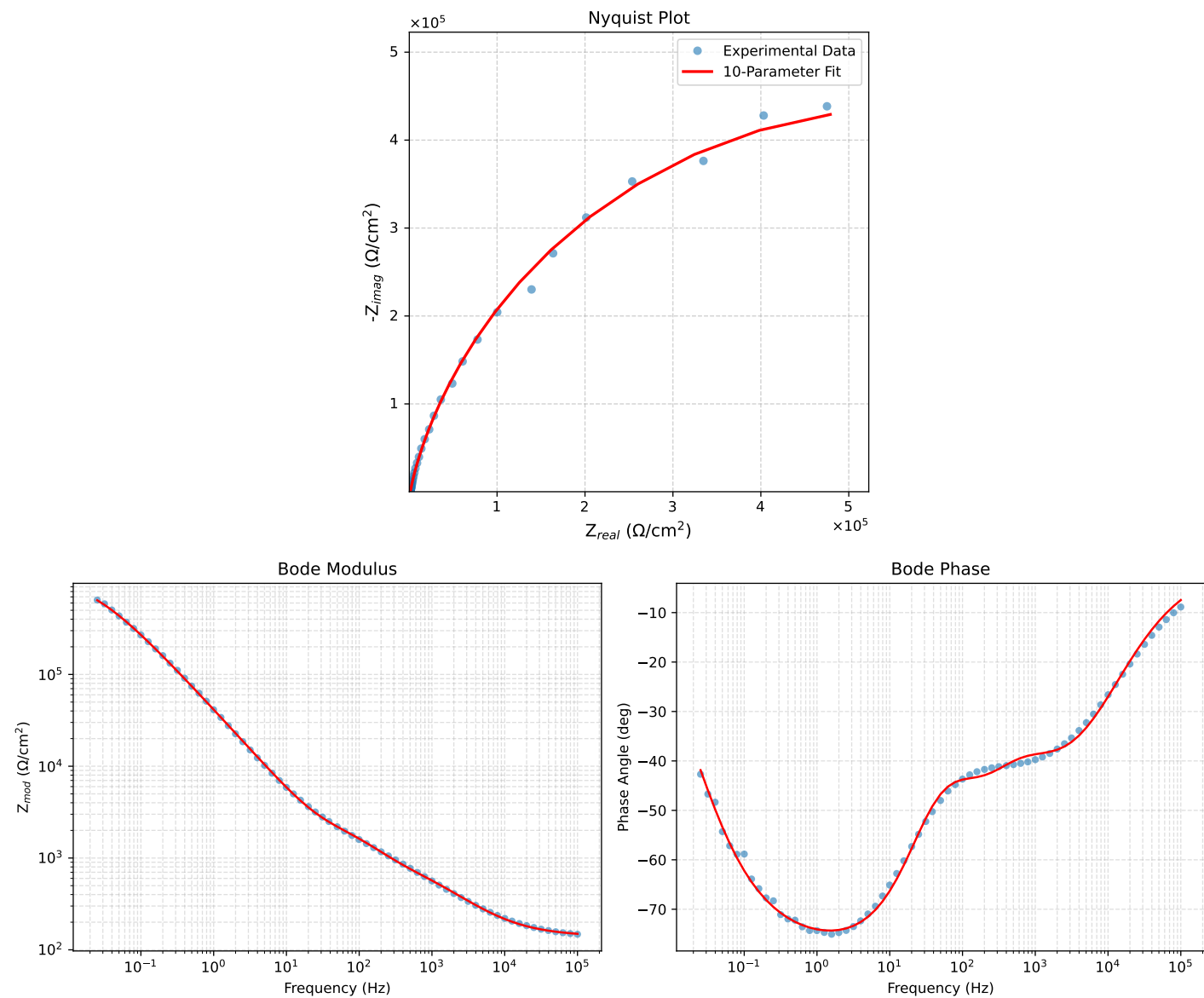


Figure 2: EIS Fit Results for Vasco_LNV3_3

Fitted Data: Vasco_LNV3_3

Frequency (Hz)	$\mathbf{z}_{real}$ (Ω)	$-\mathbf{z}_{imag}$ (Ω)	$\mathbf{z}_{mod}$ (Ω)	$\mathbf{z}_{phase}$ ($^\circ$)
1.0002e+05	1.4818e+02	1.9395e+01	1.4944e+02	-7.4570
7.9512e+04	1.4987e+02	2.2804e+01	1.5161e+02	-8.6892
6.3105e+04	1.5191e+02	2.7066e+01	1.5430e+02	-10.1025
5.0215e+04	1.5433e+02	3.1903e+01	1.5759e+02	-11.6796
3.9902e+04	1.5725e+02	3.7612e+01	1.6169e+02	-13.4511
3.1699e+04	1.6078e+02	4.4307e+01	1.6677e+02	-15.4073
2.5137e+04	1.6506e+02	5.2199e+01	1.7311e+02	-17.5496
1.9980e+04	1.7017e+02	6.1306e+01	1.8088e+02	-19.8119
1.5879e+04	1.7636e+02	7.1886e+01	1.9045e+02	-22.1763
1.2598e+04	1.8391e+02	8.4215e+01	2.0227e+02	-24.6038
1.0020e+04	1.9295e+02	9.8238e+01	2.1652e+02	-26.9827
7.9282e+03	2.0414e+02	1.1463e+02	2.3412e+02	-29.3153
6.3079e+03	2.1735e+02	1.3278e+02	2.5469e+02	-31.4213
5.0347e+03	2.3296e+02	1.5283e+02	2.7862e+02	-33.2662
3.9931e+03	2.5213e+02	1.7570e+02	3.0731e+02	-34.8714
3.1441e+03	2.7560e+02	2.0159e+02	3.4146e+02	-36.1846
2.5195e+03	3.0099e+02	2.2754e+02	3.7731e+02	-37.0886
2.0008e+03	3.3126e+02	2.5639e+02	4.1889e+02	-37.7387
1.5807e+03	3.6616e+02	2.8772e+02	4.6568e+02	-38.1597
1.2536e+03	4.0404e+02	3.2054e+02	5.1574e+02	-38.4262
1.0007e+03	4.4378e+02	3.5491e+02	5.6825e+02	-38.6506
7.9003e+02	4.8823e+02	3.9479e+02	6.2788e+02	-38.9599
6.2881e+02	5.3390e+02	4.3872e+02	6.9103e+02	-39.4106
5.0223e+02	5.8232e+02	4.8908e+02	7.6046e+02	-40.0262
4.0015e+02	6.3629e+02	5.4892e+02	8.4035e+02	-40.7841
3.1550e+02	7.0054e+02	6.2244e+02	9.3712e+02	-41.6213
2.5202e+02	7.7101e+02	7.0247e+02	1.0430e+03	-42.3369
2.0032e+02	8.5459e+02	7.9411e+02	1.1666e+03	-42.8993
1.5801e+02	9.5365e+02	8.9795e+02	1.3099e+03	-43.2768
1.2556e+02	1.0601e+03	1.0065e+03	1.4618e+03	-43.5143
9.9734e+01	1.1733e+03	1.1243e+03	1.6250e+03	-43.7794
7.9449e+01	1.2865e+03	1.2539e+03	1.7965e+03	-44.2661
6.2919e+01	1.3993e+03	1.4093e+03	1.9860e+03	-45.2025
4.9867e+01	1.5057e+03	1.5993e+03	2.1966e+03	-46.7265
3.8422e+01	1.6180e+03	1.8738e+03	2.4757e+03	-49.1906
3.1250e+01	1.7045e+03	2.1534e+03	2.7463e+03	-51.6377
2.4934e+01	1.8014e+03	2.5392e+03	3.1133e+03	-54.6464
2.0032e+01	1.9038e+03	3.0110e+03	3.5624e+03	-57.6958
1.5625e+01	2.0391e+03	3.6894e+03	4.2154e+03	-61.0703
1.2467e+01	2.1892e+03	4.4652e+03	4.9729e+03	-63.8824
9.9734e+00	2.3730e+03	5.4119e+03	5.9093e+03	-66.3240
7.9719e+00	2.6045e+03	6.5783e+03	7.0752e+03	-68.4000
6.3516e+00	2.9019e+03	8.0273e+03	8.5357e+03	-70.1247
5.0134e+00	3.2975e+03	9.8793e+03	1.0415e+04	-71.5422
3.9860e+00	3.7879e+03	1.2078e+04	1.2658e+04	-72.5880
3.1758e+00	4.4063e+03	1.4731e+04	1.5375e+04	-73.3468
2.5161e+00	5.2145e+03	1.8039e+04	1.8777e+04	-73.8768
1.9955e+00	6.2434e+03	2.2047e+04	2.2914e+04	-74.1884
1.5879e+00	7.5383e+03	2.6837e+04	2.7876e+04	-74.3104
1.2581e+00	9.2270e+03	3.2749e+04	3.4024e+04	-74.2647
9.9777e-01	1.1386e+04	3.9868e+04	4.1462e+04	-74.0609
7.9274e-01	1.4140e+04	4.8384e+04	5.0408e+04	-73.7088
6.2903e-01	1.7710e+04	5.8674e+04	6.1289e+04	-73.2043
4.9888e-01	2.2338e+04	7.1027e+04	7.4456e+04	-72.5419
3.9860e-01	2.8126e+04	8.5249e+04	8.9769e+04	-71.7409
3.1672e-01	3.5800e+04	1.0247e+05	1.0854e+05	-70.7424
2.5148e-01	4.5812e+04	1.2279e+05	1.3106e+05	-69.5397
1.9998e-01	5.8742e+04	1.4629e+05	1.5764e+05	-68.1221
1.5879e-01	7.5638e+04	1.7349e+05	1.8926e+05	-66.4438
1.2601e-01	9.7567e+04	2.0436e+05	2.2645e+05	-64.4788
1.0016e-01	1.2552e+05	2.3829e+05	2.6933e+05	-62.2209
7.9503e-02	1.6123e+05	2.7500e+05	3.1877e+05	-59.6173
6.3140e-02	2.0571e+05	3.1287e+05	3.7444e+05	-56.6747
5.0123e-02	2.6013e+05	3.5005e+05	4.3612e+05	-53.3833
3.9833e-02	3.2435e+05	3.8368e+05	5.0241e+05	-49.7900
3.1638e-02	3.9803e+05	4.1103e+05	5.7217e+05	-45.9208
2.5126e-02	4.7903e+05	4.2918e+05	6.4317e+05	-41.8582